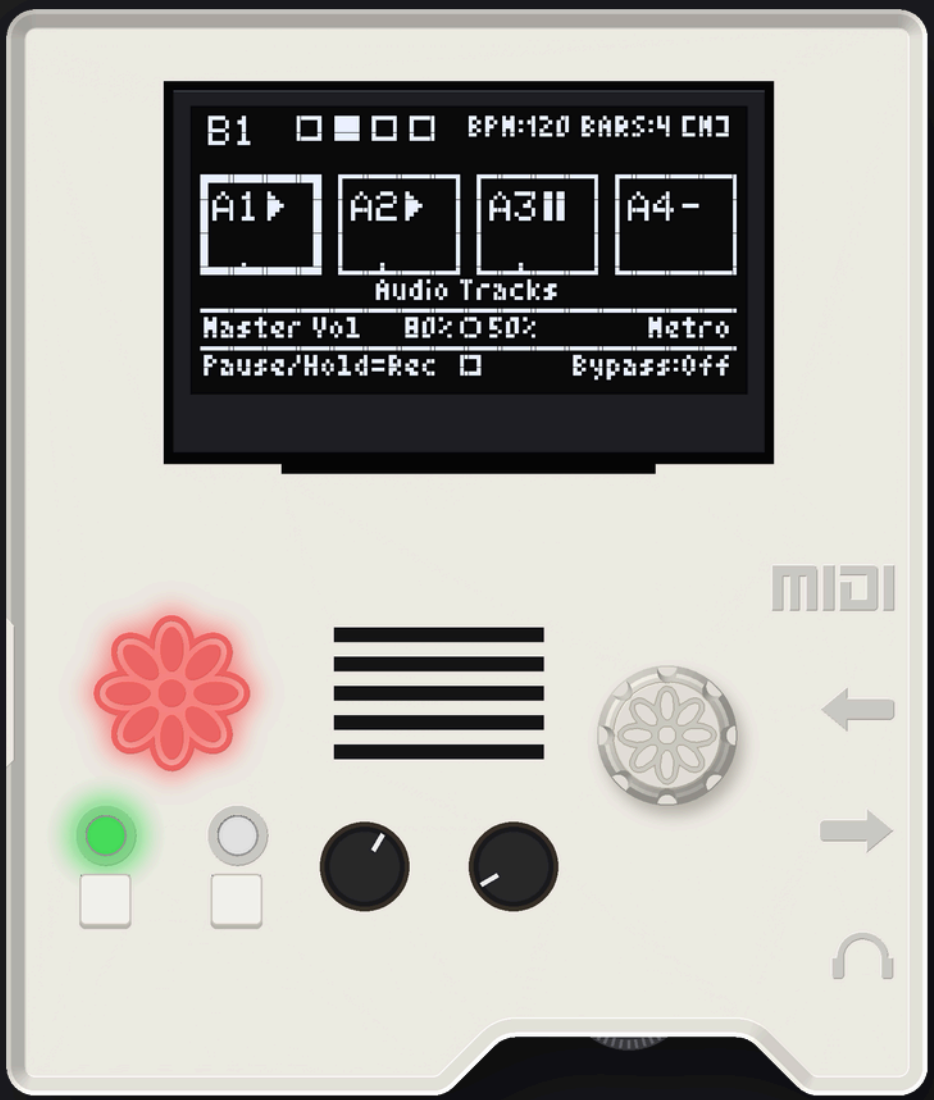
PZD

GROOVEBOX

User Manual



Eight engines · four-track looper · eight-track MIDI sequencer

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CHAPTER 1

Meet the Groovebox

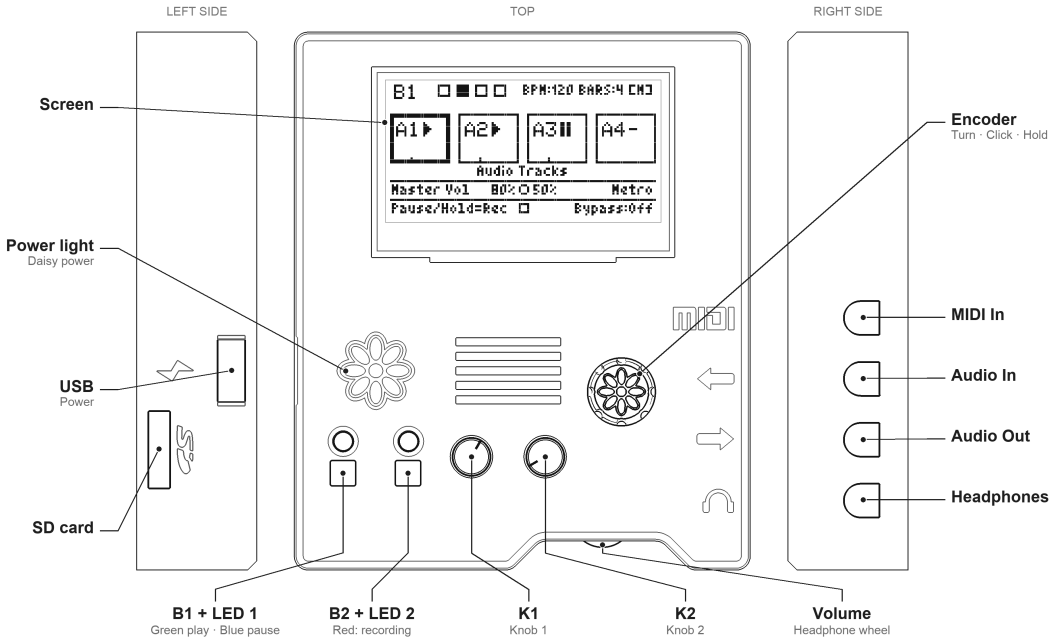
The PZD Groovebox is a pocket, multitimbral groovebox built on the Electrosmith Daisy Pod. It records and loops audio, plays eight synth and sampler engines from a MIDI keyboard, and sequences them, all at the same time on one small board. A 128 × 64 screen shows every setting, and the Pod's two knobs, two buttons and one clickable encoder control all of it.

1.1 What it does

- **A four-track looper.** Record, overdub and undo audio from the input, locked to one tempo, with a filter, drive, tremolo and effect sends on every track.
- **Eight engines.** Two DX7-style FM synths (DXD-A and DXD-B), a string machine (Ensemble), a drawbar organ, a supersaw synth, a granular instrument (Grains) and two samplers, one for melodic sounds (SMPVO) and one for drums (SMPDR). Any of them can play at once.
- **An eight-track MIDI sequencer.** Record from a keyboard or draw notes on a grid, with eight patterns per track and chains of patterns.
- **Scenes and a song.** Sixteen scenes switch patterns and audio tracks together on the beat, and a song of up to 64 steps arranges them.
- **An arpeggiator** on every engine and every MIDI track.
- **A mixer and shared effects:** a plate reverb, a stereo delay and a master filter.
- **Saving and export.** Performances, snapshots and presets save to the SD card, and anything you play exports as a WAV file.

1.2 Controls and connections

The Groovebox is a Daisy Pod in the PZD case, with a screen added. Every page is worked the same way: the encoder moves around, the two knobs set the two values on the upper row of the screen, and the two buttons do what the lower row says.



Control	What it does
Encoder	Turn to move between tracks or pages. Click to go in or back out. Hold to go Home, or from Home to Global. Click three times quickly for All Notes Off.
K1, K2	Set the two values on the screen's upper row: K1 the left one, K2 the right one.
B1, B2	Do what the screen's lower row says: B1 the left action, B2 the right one. Some actions need a tap, others a hold.
Volume wheel	The headphone level, in the notch at the bottom edge.
MIDI In	A 3.5 mm TRS MIDI input, Type A.
Audio In, Audio Out	Stereo line input and output, on 3.5 mm jacks.
Headphones	The headphone output.
USB	Power. MIDI doesn't come in over USB.
SD card	A micro SD card, FAT32, for everything you save.

1.3 The lights

Light	What it shows
LED 1	Green while any track plays, blue while paused. It flashes blue in time with the click during a count-in.
LED 2	Red while anything records: an audio track, a MIDI track, a capture, or a live export.
The daisy	The Daisy Seed's power light.

1.4 The card and the firmware

The SD card holds your performances, snapshots, presets, samples and exports. Start with the card template, **sdcard-template.zip**, attached to each release: unzip it onto the root of a FAT32 micro SD card. It holds the demo song, starter sounds and the folders the Groovebox saves into.

> **Switch off first.** Always switch off before taking the card out or putting it in.

Installing new firmware

Copy **main.bin** from the release onto the root of the card, put the card in, and switch on with a steady power supply. The Groovebox finds the file and installs it by itself.

The first time: the bootloader

The firmware starts through a small bootloader, which a Pod needs once. Any Pod that has run this firmware from v1.6.0 onward already has it. Otherwise:

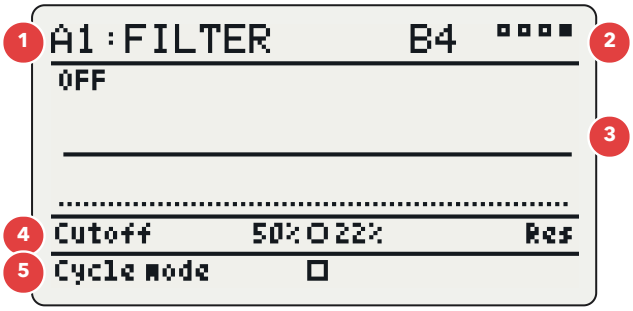
1. Put the Pod into DFU mode: hold **BOOT**, tap **RESET**, let go of **BOOT**.
2. Open Electrosmith's web programmer (flash.daisy.audio) and flash **dsy_bootloader_v6_4-intdfu-2000ms.bin**, attached to each release.
3. Then install the firmware from the card as above.

CHAPTER 2

Finding your way

Every screen on the Groovebox is laid out the same way, and the same few moves get you anywhere. This chapter covers them once, so the rest of the manual can just name the page and the control.

2.1 The screen



1. Where you are: the area, then the page. This is audio track A1's Filter page.
2. Where the loop is: the bar it's playing, and a box for each beat.
3. What the page does. Most pages draw it: here, the filter's response.
4. The knob row, marked by the circle. K1's job is on the left with its value beside the circle; K2's is on the right.
5. The button row, marked by the square. B1's job is on the left, B2's on the right.

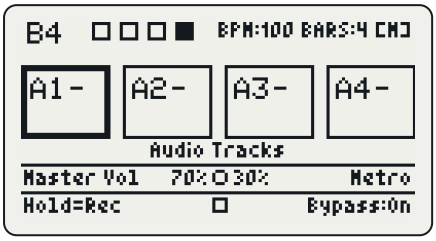
A small `*` after a page's title means its sound has changed since it was last loaded or saved.

2.2 Turning, clicking and holding

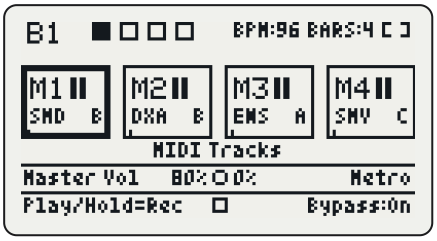
Encoder	What it does
Turn	On Home, moves through the tracks, then the Scenes and Song pages. Anywhere else, moves through that area's pages.
Click	Goes one level in, or back out. From Home it opens the selected track. On a Global page with **PUSH ENC TO OPEN**, it opens that engine or mixer, and clicking again comes back.
Hold	From Home, opens Global. From anywhere else, goes back Home.
Triple-click	Three clicks within about a second: All Notes Off on every engine, for a stuck note. You stay on the page you were on.

2.3 Home

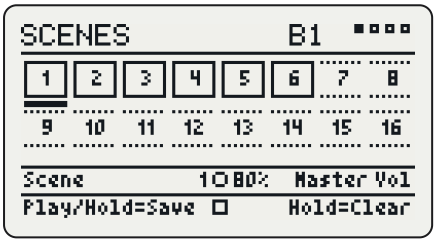
Home is where you start and where you come back to. Turning the encoder there moves through the four audio tracks (A1–A4), the eight MIDI tracks (M1–M8), then the Scenes and Song pages. The tracks show four to a page, each box with its state and a playhead along the bottom.



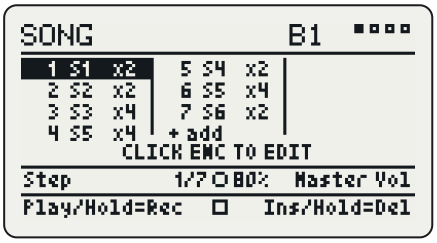
The audio tracks



The MIDI tracks



Scenes



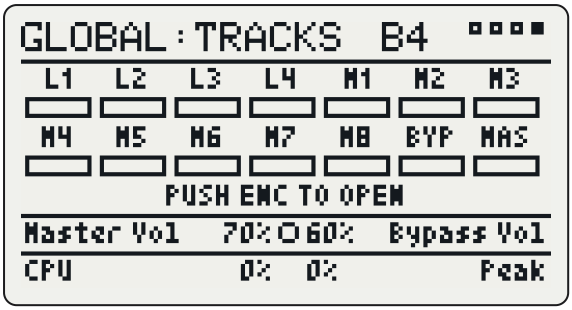
Song

On Home, K1 is the master volume and K2 the metronome's. B1 records, pauses and overdubs the selected track, and B2 switches Bypass, which lets you hear the input. Each box shows its track's state:

Sign	The track is
▶	playing
⏸	paused
●	recording
○	counting in, ready to record
-	empty

2.4 Global

Hold the encoder on Home to open Global, the settings that aren't any one track's. Turn through its pages; hold again to go back Home.



Global:Tracks, the first Global page

Page	For
Tracks, Synths	The mixer: the tracks' and the engines' levels, with Master and Bypass.
Sends	Every channel's reverb and delay send on one page.
Tempo	BPM, the loop's length in bars, and the metronome.
Perf, Snap	Saving and loading Performances and Snapshots.
SD Mgmt	Copying and deleting files on the card.
Export	Writing a WAV of a loop or the whole song.
Display	The screen saver.
Speed	Vari-speed, scrub and freeze for the whole project.
Filter	The master filter.
Reverb, RevTone, RevDiff	The shared reverb.
Dly-Rev, Delay, DlyTone	The shared delay, and how much of it goes into the reverb.
Bypass	The monitored input's sends and level.
MIDI	Every engine's MIDI channel, and the sustain pedal.
Arp	Every engine's arpeggiator.
Grains … SMPDR	One page per engine: switch it on, and click to open it.
Looper	Switches the whole looper on or off.

2.5 Knobs pick up

The knobs don't jump the moment you touch them. Each value on the screen is the setting as it is now; turning a knob only takes control once it reaches that value. So moving to a new page never changes anything until you mean it to.

2.6 The screen saver

After a few minutes with nothing touched, the screen dims, then switches off a minute later. The first touch only wakes it, so it can't change anything by accident. Set the time on **Global:Display** : Off, 1, 2, 5, 10 or 30 minutes (5 to start with). B1 there puts it to sleep now, and B2 saves the time for every power-on.

CHAPTER 3

Audio tracks

The four audio tracks, A1–A4, are the looper. Each records the audio input (and any engine you play while it records) into a loop that all four share, locked to one tempo. The loop's length is set in bars on **Global:Tempo**.

3.1 Recording a loop

1. On Home, turn to the track you want, A1–A4.
2. Hold B1. If the metronome is on, it counts in one bar first, with LED 1 flashing blue.
3. Play. LED 2 is red while it records, and it stops by itself at the end of the loop. Tap B1 to stop early.

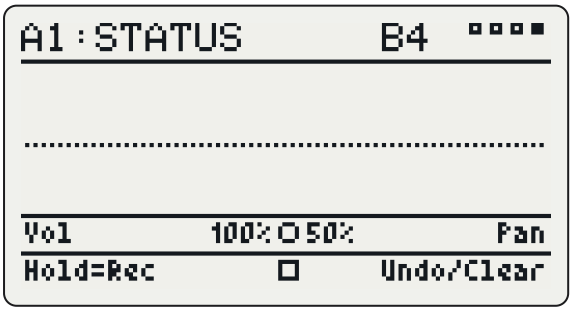
Then:

On Home	What happens
Tap B1	Pause or play the selected track.
Hold B1	Overdub: record more on top of the track, from the next count-in.
K1	Master volume.
K2	Metronome volume.
B2	Bypass on or off: hear the input without recording it.

> **Engines record too.** Anything you play on an engine while a track records goes into the take, mixed with the input. No cable needed.

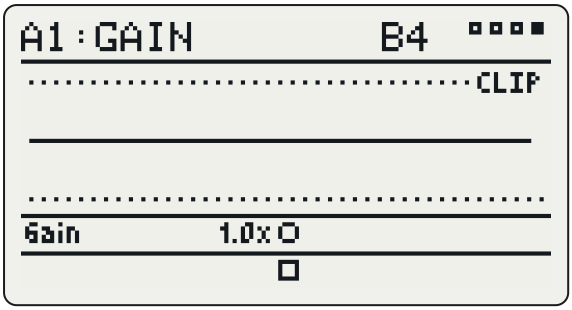
3.2 A track's pages

Click the encoder on a selected track to open its pages, and turn to move between them. Click again, or hold, to go back.



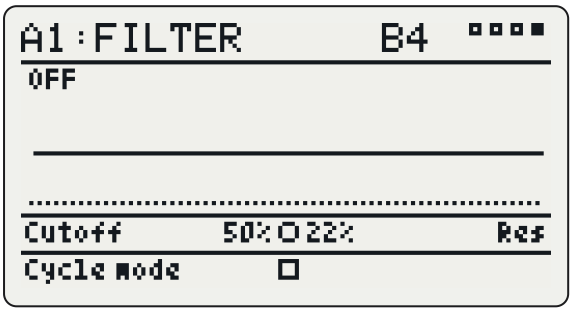
Status

The track's waveform, with a moving playhead. K1 is its volume and K2 its pan. B1 records, pauses and overdubs, as on Home. Tap B2 to undo the last overdub; hold it for about a second to clear the track.



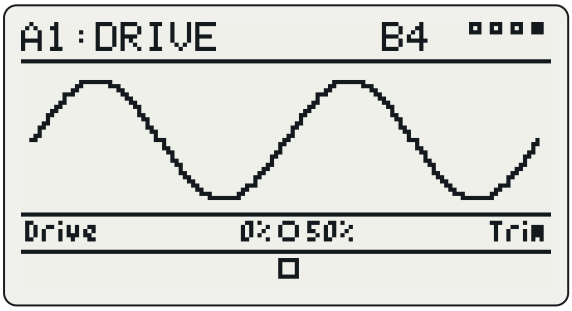
Gain

K1 sets the input gain, 1× to 4×. While this page is open you hear the input through it, and its level shows against the clip line: set it as high as it goes without touching the line.



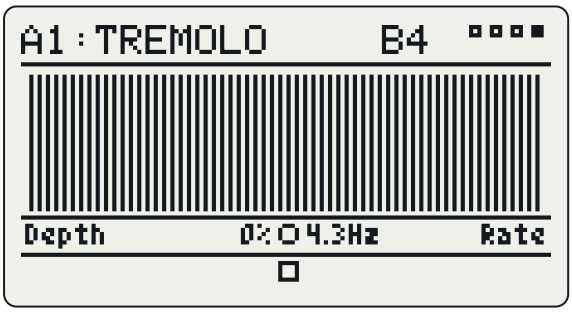
Filter

B1 picks the filter: Off, low-pass, high-pass or band-pass. K1 is the cutoff and K2 the resonance. The page draws the filter's response.



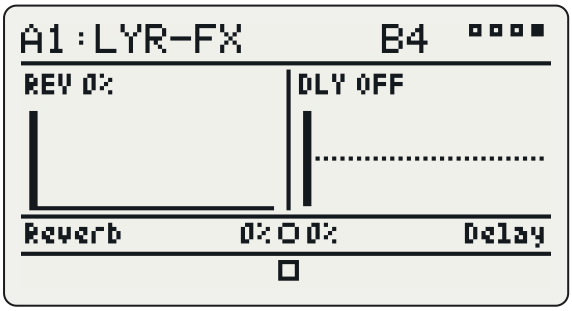
Drive

K1 drives the track into saturation (0 is off). K2 trims the level back, with the middle leaving it as it is. The page draws the driven wave.



Tremolo

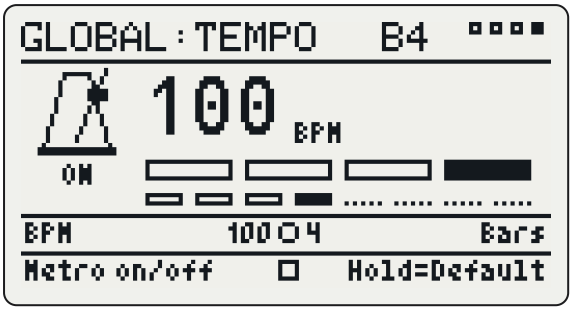
K1 is the depth (0 is off) and K2 the rate, 0.5 to 10 Hz. The page draws the level over time.



LYR-FX

K1 sends the track to the shared reverb and K2 to the shared delay. The page draws both, small.

3.3 Tempo and the loop's length



On **Global:Tempo**, K1 sets the tempo, 50 to 240 BPM, and K2 the loop's length, 1 to 16 bars. At slow tempos the longest loops aren't available, because a loop has a fixed amount of memory. B1 switches the metronome on and off. Holding B2 saves the power-on defaults: the tempo and loop length, with the master volume, Bypass, the master filter, the reverb and the delay.

The page shows the BPM large, a metronome whose arm swings on each beat while the click is on, a box for each beat, and a box for each bar of the loop. The bar playing is filled; dotted bars are ones you could still add. While any track holds audio, the page shows **LOCKED**: the tempo and length are fixed until the tracks are cleared.

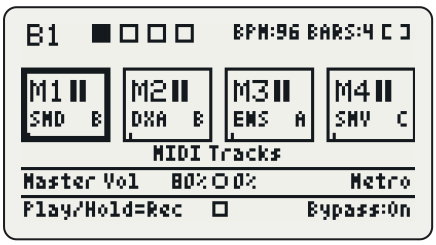
3.4 Switching the looper off

Global:Looper switches all four tracks on or off with B1, freeing their share of the processor for the engines. B2 sets whether the looper is on at power-on, and clicking the encoder there opens Home.

CHAPTER 4

MIDI tracks

The eight MIDI tracks, M1–M8, record what you play on a keyboard and play it back through any of the eight engines. They sit on Home after the audio tracks and work the same way: hold B1 to record a pass, tap to pause, hold again to overdub. Each has eight patterns, a chain to play them in order, quantise and swing, and a grid for drawing and editing notes.



Home: each box shows the track's engine and the pattern playing

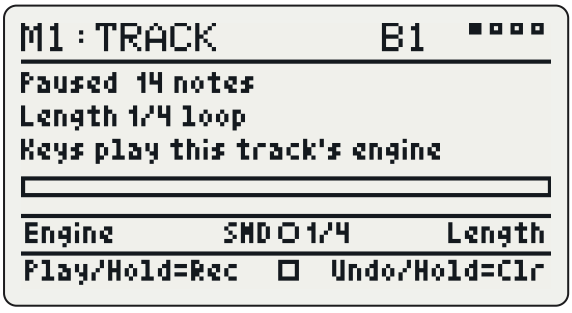
4.1 Recording a MIDI track

1. On Home, turn past A4 to a MIDI track. While it's selected, the keyboard plays its engine.
2. Hold B1. After the count-in, play: the track records one pass of its length, then loops it.
3. Tap B1 to pause or play. Hold B1 again to overdub more notes on top.

The engine has to be switched on (on its Global page) for you to hear it. A track's notes are saved with the Performance.

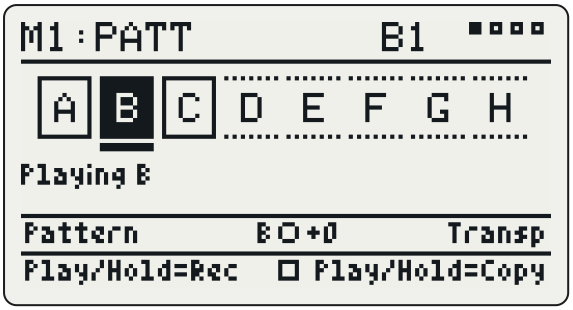
4.2 A track's pages

Click the encoder on a MIDI track to open its pages, and turn to move between them. A click goes back to Home. B1 records, pauses and overdubs on every page, as on Home.



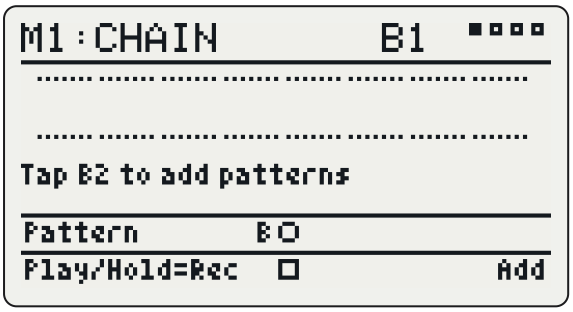
Track

K1 picks the engine the track plays. K2 sets its length: ¼, ½, 1, 2 or 4 times the loop, up to 16 bars, so a one-bar beat can repeat under a four-bar loop. Tap B2 to undo the last pass; hold it to clear the track.



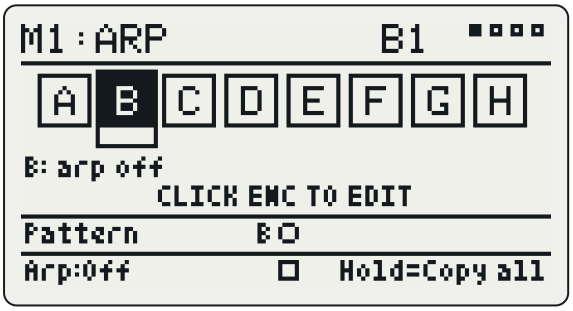
Patt

Each track has eight patterns, A–H, each with its own notes, length, quantise, swing and transpose. K1 picks a pattern and K2 transposes it, ±24 semitones. Tap B2 to play the picked pattern: it takes over when the one playing reaches its end. Hold B2 to copy the playing pattern into the picked one, to start a variation.



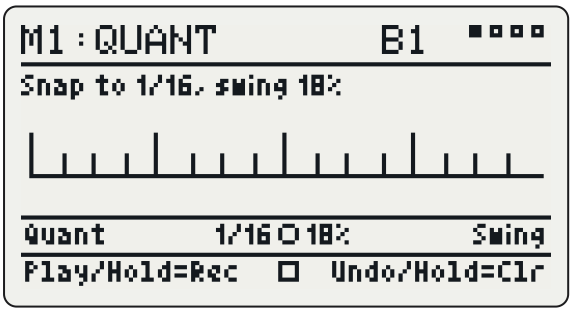
Chain

A chain plays up to eight patterns in order, each for its own length: A A A B is three bars of A, then a fill. K1 picks a pattern and a B2 tap adds it to the chain; hold B2 to clear the chain. A chain starts with the loop.



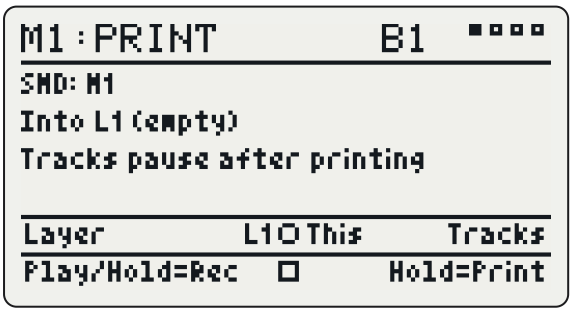
Arp

Every pattern has its own arpeggiator. K1 picks a pattern and B1 switches its arp on or off; click the encoder to edit it (see Chapter 6). Hold B2 to copy the picked pattern's arp to every pattern.



Quant

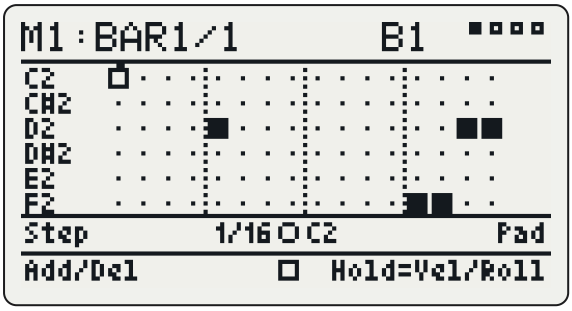
K1 sets the quantise, from Off through 1/32T to 1/4, and K2 the swing, up to 100%. Both apply on playback: the notes you recorded never move, so you can change your mind.



Print

Records the track into an audio track, to free the engine or keep a take. K1 picks the audio track; K2 picks This track or every MIDI track (Tracks). Hold B2 to print, and tap it to stop. The MIDI tracks pause afterwards. Past the audio tracks, K1 reaches **Notes**: holding B2 then writes the arp's notes into the track's next empty pattern as real, editable notes.

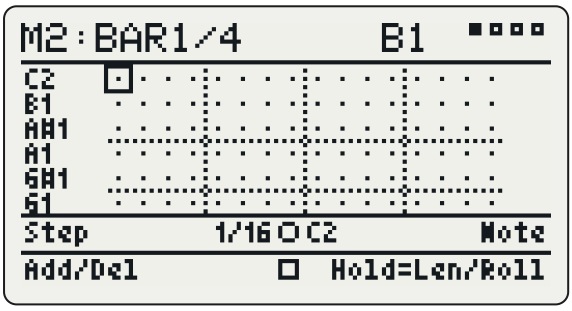
4.3 The grid



A drum grid: M1 plays SMPDR

After Print come the grid pages, one per bar. Most engines get a piano roll, C1 to C6; a track playing SMPDR gets a drum grid instead, with one row per pad from C2.

K1 moves along the steps and K2 picks the note. B1 adds or deletes a note, which sounds as it's added. Hold B2 to set the note's velocity (on a drum grid) or length with K1, and its roll with K2: ×2 to ×8 quick repeats inside one step. With K1 fully left, an edit applies to the whole row.



A piano roll: M2 plays DXD-A

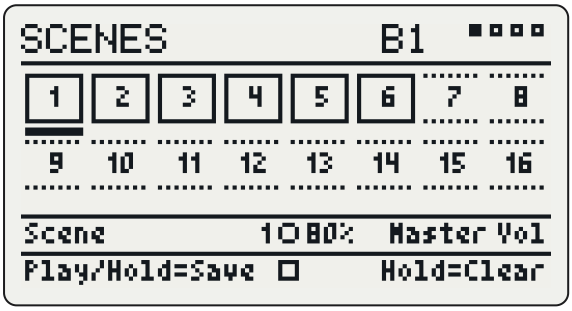
If the track's arp is on, its notes show over the keys you recorded: a hollow bar where it plays a note, a line through a recorded key it plays.

CHAPTER 5

Scenes and Song

Scenes remember what plays together, and the song plays scenes in order. Together they turn a set of loops and patterns into a finished piece, which exports as one WAV file.

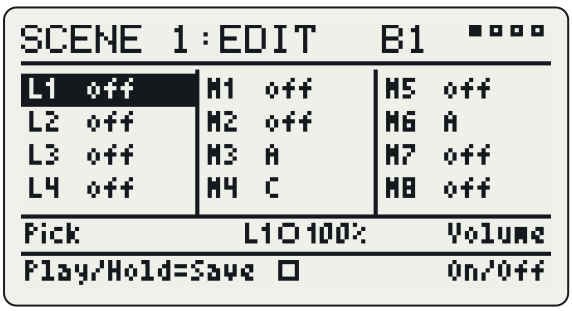
5.1 Scenes



There are sixteen scenes. Each remembers every MIDI track's pattern or chain (or off), and every audio track on or off at its own volume. The Scenes page comes after M8 on Home.

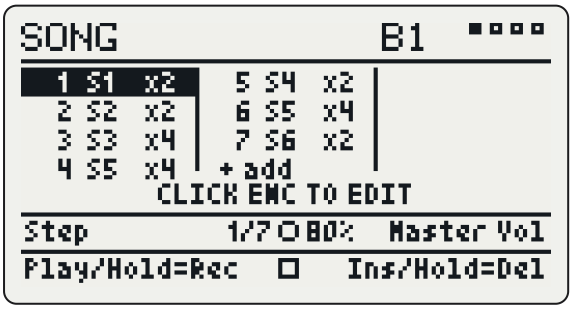
Get the tracks playing as a section should sound, then hold B1 to save it into the scene K1 has picked. Tap B1 to launch a scene: everything switches together on the loop's next downbeat, every track from its top. Tap again before then to cancel. Hold B2 to clear a scene. K2 is the master volume.

Editing a scene



Click the encoder on Scenes to edit the picked scene directly. K1 picks a track; K2 sets its volume (audio tracks) or its pattern, Off or A–H (MIDI tracks). B2 switches the track on or off in the scene. B1 launches and saves as on the Scenes page.

5.2 The song



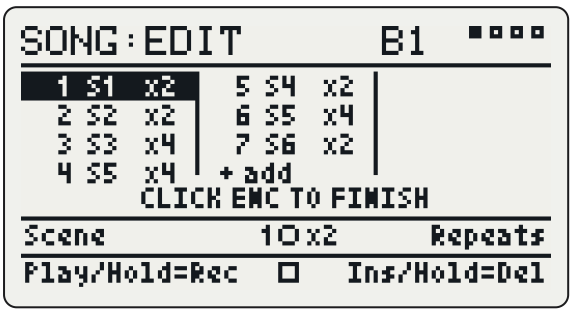
A song is up to 64 steps, each a scene played 1 to 32 times. The Song page comes after Scenes. K1 picks a step and K2 is the master volume.

Tap B1 to play from the picked step, and again to stop. While it plays, the list follows the step playing, and at the end everything stops.

Building a song

Two ways, which mix freely:

- **Record it.** Hold B1 on the Song page, then launch scenes on the Scenes page: each launch becomes a step, for as many loops as it played.
- **Write it.** Tap B2 to insert a copy of the picked step, and hold B2 to delete one. Click the encoder to edit the picked step: K1 sets its scene and K2 its repeats.



Editing a step

Click again to stop editing. The patterns, chains, scenes and song all save in the Performance, with the audio and every engine's sound.

5.3 Exporting the song

On `Global:Export`, turn K1 to **Song** and tap B1. The song plays from the step picked on the Song page to its end, then until the effects have faded, and is written to the card as one WAV of any length. The outputs are muted while it runs. About every 37 seconds it pauses for a moment to write to the card, so the file has no gaps. See Chapter 14 for the other export modes.

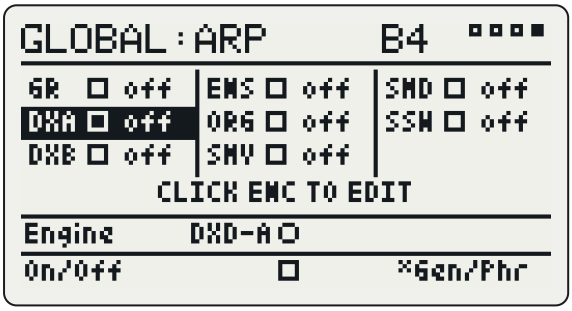
CHAPTER 6

The arpeggiator

Every engine has its own arpeggiator for playing live, and every MIDI track has one too, with its own settings for each pattern. Any number can run at once: DXD-A arping sixteenths while the Organ arps eighths.

Press one key and the arp builds a chord on it from its key and scale, and plays a pattern over it; press another key and the pattern moves there. **Hold a chord**, two or more keys, and it plays that chord instead.

6.1 Global:Arp



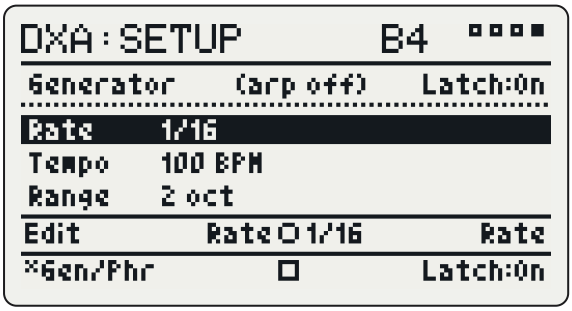
Every engine's arp on one page. K1 picks an engine and B1 switches its arp on or off. B2 switches it between Generator and Phrase. Click the encoder to edit the picked arp.

A MIDI track's arps are on the track's own Arp page (Chapter 4). Either way, the arp opens on the same pages, below; a click goes back to where you opened it.

6.2 Lock or Follow: what one key means

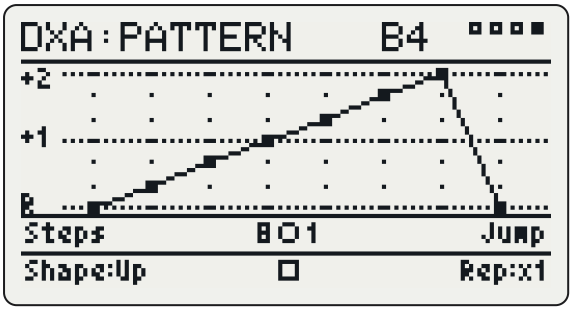
- **Lock** keeps the key put. In C major, C plays C major, D plays D minor, F plays F major and B plays B diminished: every key fits the song. A held chord's notes are kept in the scale.
- **Follow** makes the key pressed the key, so every key plays the same kind of chord. A held chord is recognised in any inversion: major, minor, 7ths, diminished, augmented, sus, 6ths and add9.

6.3 The arp's pages



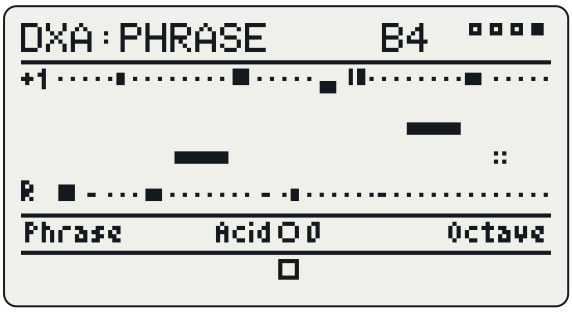
Setup

K1 picks **Rate**, **Tempo** or **Range**, and K2 sets it. Rate is 1/4 to 1/32, with triplets. Tempo is the same as **Global:Tempo** . Range spreads the chord over 1 to 3 octaves. B1 switches between Generator and Phrase. B2 switches **Latch**: the arp keeps playing after you let go, and only pressing keys changes the chord.



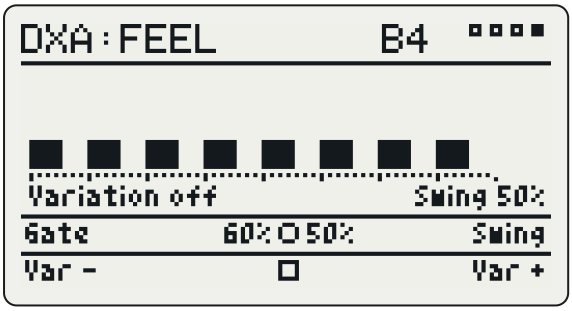
Pattern (Generator)

Generator walks the chord's notes. K1 sets the **Steps**, 1 to 16, and K2 the **Jump**, 1 to 7 notes at a time. B1 picks the next **Shape**: Up, Down, Up/Dn, 3+2-, 2+1-, Zigzag, In/Out, Pedal or Scatter. B2 sets **Repeat**, each note ×1 to ×4. The page draws the notes as a ladder and the walk through them.



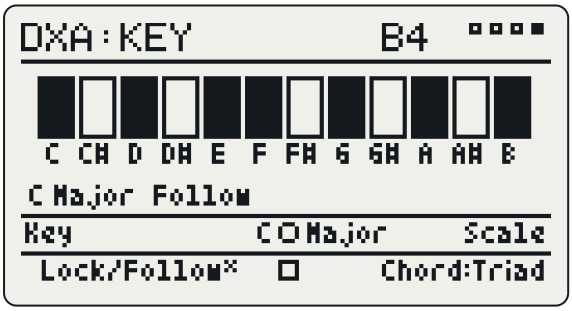
Phrase

Phrase plays one of 24 ready-made phrases from the key pressed: basslines, arps, riffs and melodies. They're written in scale steps, so they play in the arp's scale, and every step has its own velocity, length, octave, chance and ratchet. K1 picks the phrase and K2 moves it up or down two octaves. The page draws it as a piano roll.



Feel

K1 is the **Gate**, how long each note plays, and K2 the **Swing**: 50% straight, 66% triplet, up to 75%. B1 and B2 step the **Variation** down and up; hold them to scroll quickly. Variation 0 is straight, and 1 to 99 each add a fixed set of accents, short and long notes, octave jumps, rests and ratchets, the same every time.



Key

K1 sets the key and K2 the scale: Major, Minor, Dorian, Mixolydian, Harmonic Minor, Pentatonic Major and Minor, or Blues. B1 switches Lock and Follow. B2 picks the next **Chord**: Triad, 7th, Sus2, Sus4, Power (root and fifth) or the whole Scale, for runs.

6.4 Playing in time

The arp starts on the next step after you play, and a new chord takes over at the next step, so the note playing is never cut off and a chord's notes don't have to land together. Arps follow the loop while the looper runs, and the tempo when it doesn't. They stop and start with the looper, on the beat.

6.5 Arps on MIDI tracks

While a MIDI track is selected, the keyboard plays through the track's arp, and the track records the keys you press. On playback they go through the arp again, so its Shape, Scale or Variation can still change

after recording. Each pattern has its own arp: copying a pattern copies its arp. The arp starts its pattern over at the top of the track, so every loop plays the same.

To edit the arp's notes one by one, turn them into real notes with **Print → Notes** (Chapter 4).

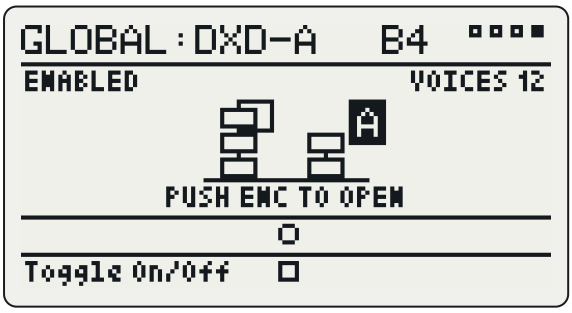
CHAPTER 7

The engines

The Groovebox has eight engines, and any of them can play at once, each on its own MIDI channel. They play from a keyboard, from the MIDI tracks and from the arpeggiators, and anything they play while an audio track records goes into the take.

Engine	Global page	What it is
Dexed	DXD-A, DXD-B	Two DX7-style FM synths, to layer or split: electric pianos, bells, basses, brass. Chapter 8.
Ensemble	Ensmbl	A string machine with a built-in chorus: strings and pads. Chapter 9.
Organ	Organ	A tonewheel-style drawbar organ, with percussion and key click. Chapter 9.
Supersaw	SSaw	Seven detuned saws in the JP-8000 style: leads, plucks, big chords. Chapter 10.
Grains	Grains	A granular instrument: textures and pads from audio you capture or import. Chapter 11.
Sample Sound	SMPVO	One captured or imported sound, or a multi-sample, played across the keyboard. Chapter 12.
Sample Drum	SMPDR	Drum kits and breaks on pads. Chapter 12.

7.1 Switching an engine on



Every engine starts switched off. On its Global page, B1 switches it on or off; the page shows ENABLED or DISABLED, and Dexed also shows its voices. Click the encoder to open the engine's pages; a click on any of them comes back here.

An engine that's on but silent costs almost nothing, and one that's off costs nothing at all, so switch off what a song doesn't use.

7.2 Pages every engine has

Besides its own pages, every engine has these:

Page	What it does
Filter	K1 cutoff, K2 resonance, B1 picks the filter type. The page draws the response.

Mix	The engine's output level and its **Gain**, 0 to +12 dB before its limiter, with a meter. Turn Gain up until the meter sits high without lighting the clip cap at the top, then set how loud the engine sits in the mix with Level.
FX	Its reverb and delay sends, and the shared reverb's size and delay's time. B1 points the knobs at the reverb, B2 at the delay.
MIDI	K1 is its MIDI channel, Omni or 1–16. Engines that play chords set their voices here too.
Preset	The sounds that come with it, load only.
User	Your own saves.

The Mix page

On the Ensemble, the Organ and the Supersaw, K1 is Level and K2 is Gain; pan is on the mixer. On Grains and the samplers, K1 and K2 are Level and Pan: B2 points K1 at Gain, and B1 back at Level. On Dexed, B1 picks which bar the knobs set, and B2 switches K1 to Gain and back.

Preset and User

On a **Preset** page, tap B2 to open the list, turn a knob to scroll and tap B2 to load. The list stays open after a load, so you can play each sound in turn. Where the sounds are in folders, K1 picks the folder and K2 the sound; B1 goes back a level.

On a **User** page, B1 saves: K1 picks **Overwrite** (the save you loaded) or **Save New**, and holding B2 confirms. B2 opens your saves, starting with **New** (a fresh sound); K2 scrolls and a tap loads.

A User save keeps everything on the engine's pages: the sound, its Mix settings and sends, the mod wheel target and its MIDI channel. So one save can bring back "Dexed on channel 1" and another "a drum kit on channel 10". A factory sound leaves the channel as it is.

7.3 MIDI channels

Every engine starts on channel 1, except SMPDR on 10. Set each engine's channel on its MIDI page, or all of them on **Global:MIDI**, where holding B2 saves them for every power-on (Chapter 15). Give engines different channels to play them as separate parts; give them the same one to layer them.

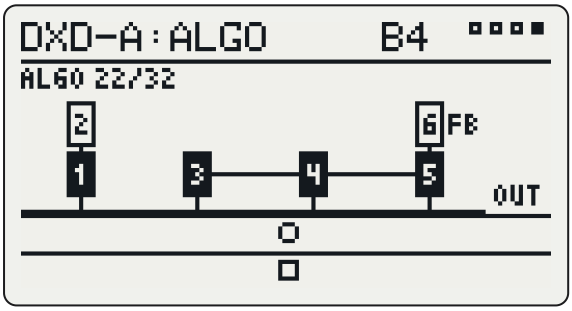
CHAPTER 8

Dexed

Dexed is a six-operator FM synth in the style of the Yamaha DX7, built on the msfa emulation core, with 3129 factory sounds in 54 folders. There are two of it, **DXD-A** and **DXD-B**, which can layer two sounds or split the keyboard between them, like a DX5. Each has its own sound, voices and User save; they share one Dexed bus with its filter, effects and fold.

8.1 Dexed's pages

Open DXD-A or DXD-B from its Global page; the pages' titles say which you're editing. Pages marked *bus* are shared: the same settings whichever engine you came in from.



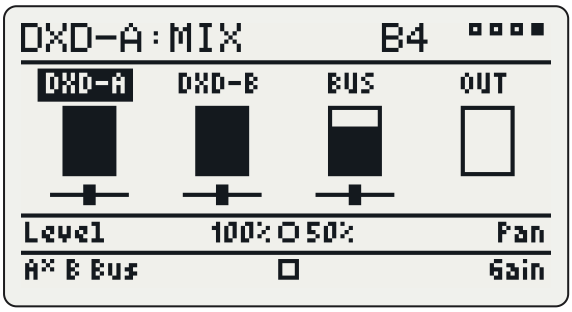
Algo

The loaded sound's algorithm, drawn like the DX7's own: operator 1 on the left, every carrier on the row above OUT, filled boxes are carriers, outlined ones modulators, and FB marks the feedback. Set the algorithm in the Advanced editor.

Page	K1	K2	B1	B2
Feedback	Feedback, 0–7			
Vibrato	LFO speed	Pitch-mod depth		
Brightness	Modulator level (centre = as saved)			
EnvSpeed	Envelope rates (centre = as saved)			
Filter *bus*	Cutoff	Resonance	Filter type	
Mix	Level / Gain	Pan	DXD-A → DXD-B → Bus	Knobs → Gain
DX-FX *bus*	Reverb / delay send	Reverb size / delay time	Reverb	Delay
DX-FOLD *bus*	Fold amount (0 = off)	Output trim		
MIDI	MIDI channel	Voices, 1–16	Lock on/off	
Split	Split point, C#1–F5	Octave shift, ±2	Layer / Split	K2 → A or B
Advanced	Click to open the full DX7 voice editor (below).			

Import		Pick a `.syx` file		Hold: import it
Preset	Roms, Dexed, Imports, then folders	Sounds in the folder	Back	Open / load
User	Your saves, as on every engine (Chapter 7).			

8.2 Two engines: Mix, MIDI and Split

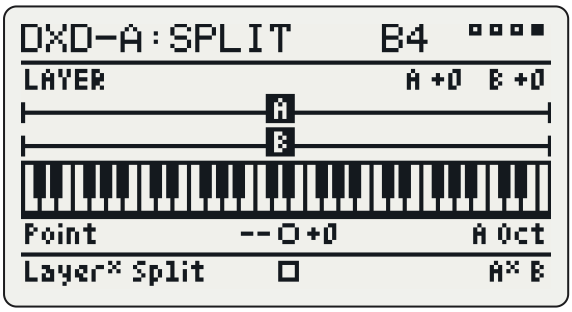


Mix

Four bars: DXD-A and DXD-B (each engine's level and pan into the bus), BUS, and OUT, the live output meter. B1 picks which of the first three the knobs set. A and B start at 100% and centre.

MIDI. With Lock on (the default), both engines play on one channel and K1 sets it for both. With Lock off, each engine's MIDI page sets its own channel, for two parts from a sequencer or two keyboards.

Voices. One engine alone can use up to 16 voices. With both on they share 24: the engine already playing keeps its voices, the one switched on gets what's left, and K2 stops at what the other leaves free.

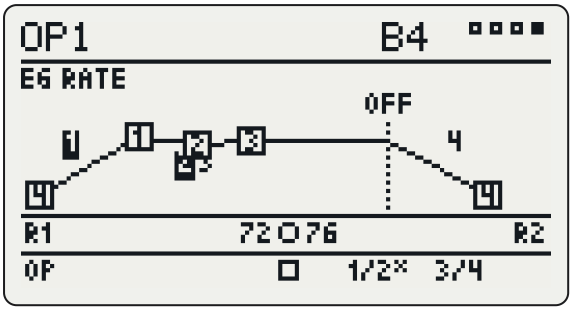


Split

In Layer, both engines play every key. In Split, DXD-A plays below the split point and DXD-B from it upwards; the keyboard drawing shows each engine's zone. Each engine has an octave shift, which works in Layer too: the same sound an octave apart thickens it. The split is for the keyboard: a MIDI track plays its engine across the whole range.

Loading a preset into one engine while the other is on changes only that engine's sound, not the shared bus, so choosing B's sound doesn't reset A's filter or mix.

8.3 The Advanced editor



Click on the Advanced page to open the full DX7 voice, with the DX7's own parameter names. The first ten pages edit one operator: B1 picks OP1–OP6, and the title shows which (with OFF when it's switched off). The last six are voice pages for the whole sound. A click goes back to Advanced.

The envelope pages draw the envelope like the DX7's own picture: L1–L4 are numbered squares and R1–R4 the slopes between them. The line starts at L4, holds at L3 until the dotted OFF line (key release), and R4 falls back to L4. What the knobs change is highlighted.

Page	K1	K2	B2
ALGORITHM	ALGORITHM, 1–32		
OSC FREQUENCY	F COARSE	F FINE	OSC MODE: RATIO / FIXED
OSC DETUNE	DETUNE, ±7		
OUTPUT LEVEL	OUTPUT LEVEL, 0–99		Operator on/off
EG RATE	R1 (or R3)	R2 (or R4)	R1/R2 ↔ R3/R4
EG LEVEL	L1 (or L3)	L2 (or L4)	L1/L2 ↔ L3/L4
LEVEL SCALING CURVE	L CURVE (or BREAK POINT)	R CURVE	K1: L CURVE ↔ BREAK POINT
LEVEL SCALING DEPTH	L DEPTH	R DEPTH	
RATE SCALING	RATE SCALING, 0–7		
SENSITIVITY	KEY VELOCITY, 0–7	A MOD SENS, 0–3	
PITCH EG RATE	R1 (or R3)	R2 (or R4)	R1/R2 ↔ R3/R4
PITCH EG LEVEL	L1 (or L3)	L2 (or L4)	L1/L2 ↔ L3/L4
LFO	LFO SPEED	LFO DELAY	LFO WAVE
LFO DEPTH	PM DEPTH	AM DEPTH	LFO KEY SYNC
P MOD SENS	P MOD SENS, 0–7		OSC KEY SYNC
KEY TRANSPOSE	TRANSPOSE (middle C = C3)		

A switched-off operator shows as a dotted box in the algorithm, and is saved in User saves and snapshots.

8.4 Your own DX7 sounds

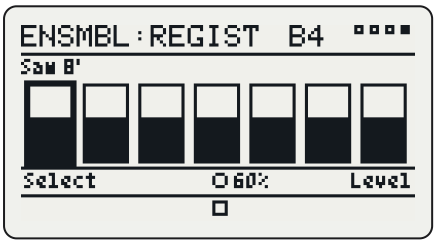
Copy DX7 SysEx banks (**.syx**) into **DEXED IMPORT/** on the card. On the Import page, K2 picks a file and holding B2 imports it: it becomes a folder under Preset → Imports. User saves are named after the sound; a second save with the same name is numbered, so copies stay apart.

CHAPTER 9

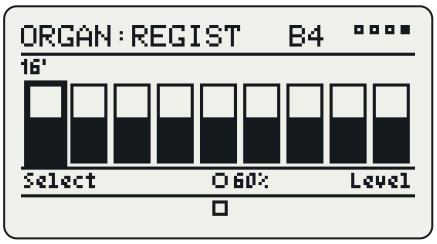
Ensemble and Organ

Two engines share one set of pages. The **Ensemble** is a divide-down string machine in the ARP Solina tradition: seven drawbars of saws and squares, a built-in stereo chorus, an LFO and velocity, with 19 factory sounds. The **Organ** is a tonewheel-style drawbar organ: nine drawbars of pure sines in Hammond order, with percussion and key click, and eight classic registrations. Open them from **Global:Ensmbl** and **Global:Organ**.

9.1 Registration



The Ensemble's seven drawbars



The Organ's nine

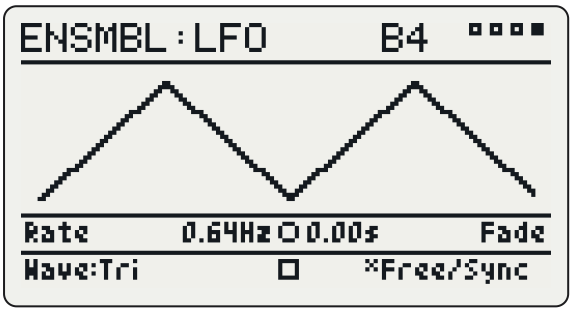
K1 picks a drawbar and K2 sets its level; the bar's footage shows at the top. The Organ's run 16', 5 1/3', 8', 4', 2 2/3', 2', 1 3/5', 1 1/3' and 1'. Both start with every drawbar at 60%.

9.2 The pages

Page	K1	K2	B1	B2
Perc *(Organ)*	Decay, about 0.15–1.5 s	Volume	Percussion on/off	2nd / 3rd harmonic
Click *(Organ)*	Key click (0 = off)			
Env	Attack / Sustain	Decay / Release		AD ↔ SR
Ensemble	Chorus depth	Chorus rate		
LFO *(Ensemble)*	Rate (a note value when synced)	Fade, 0–5 s	Wave: Tri, Saw, Square, S&H	Free / Sync
LFODep *(Ensemble)*	Pitch, Filter or Amp	Its depth (centre = off)		
Filter	Cutoff	Resonance	Filter type	
Mix	Level	Gain, 0 to +12 dB		

FX	Reverb / delay send	Reverb size / delay time	Reverb	Delay
MIDI	MIDI channel / Velocity	Voices, 1–16	Mod wheel target	K1 → channel or velocity *(Ensemble)*
Preset, User	As on every engine (Chapter 7).			

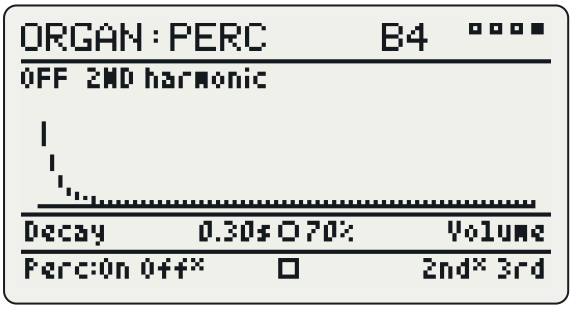
9.3 The Ensemble's LFO and velocity



The **LFO** moves the pitch of each note (with Fade, so vibrato comes in after the note starts), the filter (when its type isn't Off), or the level: a Square synced to 1/16 is a trance gate. Synced, it stays on the beat with the looper. The LFODep page sets how much it moves each one; either side of centre moves it the opposite way.

Velocity (on the MIDI page, B2 then K1): at 0 every note plays alike, as on a string machine. Turned up, softer notes are quieter, darker and slower to swell; harder ones play the registration as set, with a snappier attack.

9.4 The Organ's percussion and click



Percussion is single-trigger, as on a tonewheel organ: one shared ping restarts only when you press a key with no other key held, and every note sounding while it fades carries it. So a struck chord pings on every note, and legato lines have none. Key click adds the contact click at each note's start.

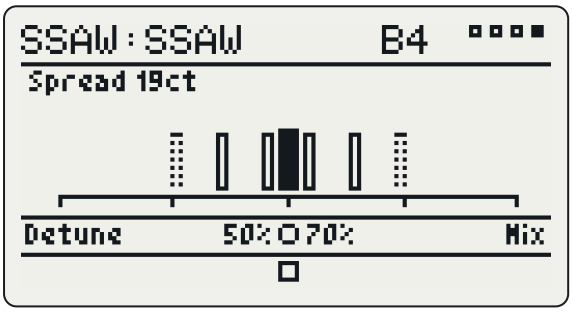
The Organ has no LFO pages and doesn't respond to velocity. Its User saves go in **ORGAN/**.

CHAPTER 10

Supersaw

The Supersaw is modelled on the Roland JP-8000's Super Saw voice: seven detuned saws with the JP's own detune spacing, knob curve and mix. Around them are a second oscillator, a resonant filter per voice with its own envelope, an LFO, mono and legato modes with portamento, velocity, and a stereo chorus. It has 42 factory sounds, from trance leads and plucks to future-bass chords and synthwave brass. Open it from **Global:SSaw**.

10.1 The SSaw page



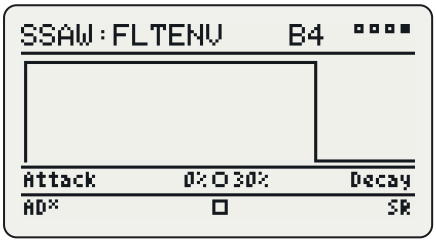
K1 is **Detune** and K2 is **Mix**. The page draws the seven saws as bars: they spread out as you turn Detune, with the JP's uneven spacing, and their height follows Mix. The centre saw is solid and the outer pair dotted.

Spread shows how far the outer saws are from the centre one, in cents. The JP's Detune climbs slowly, then steeply at the top, to ±202 cents at 100%. Mix raises the six side saws from almost silent to louder than the centre one.

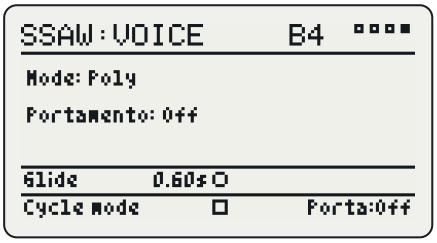
10.2 The pages

Page	K1	K2	B1	B2
Osc2	Coarse, ±24 semitones	Fine, ±50 cents	Wave: Saw, Square, Tri	
Balance	OSC1 ↔ OSC2 (centre = both full)			
Filter	Cutoff	Resonance	LPF, HPF, BPF	12 / 24 dB
FltEnv	Attack / Sustain	Decay / Release	Knobs → A + D	Knobs → S + R
FltMod	Envelope depth (centre = none)	Key follow		
AmpEnv	Attack / Sustain	Decay / Release	Knobs → A + D	Knobs → S + R
LFO	Rate (a note value when synced)	Fade, 0–5 s	Wave: Tri, Saw, Square, S&H	Free / Sync
LFODep	Pitch, Filter or Amp	Its depth (centre = off)		

Voice	Portamento time		Poly, Mono, Legato	Porta: On / Off
Chorus	Depth	Rate		
Mix	Level	Gain, 0 to +12 dB		
FX	Reverb / delay send	Reverb size / delay time	Reverb	Delay
MIDI	MIDI channel / Velocity	Voices, 1–16	Mod wheel: Off, Detune, Cutoff, Level	K1 → channel or velocity
Preset, User	As on every engine (Chapter 7).			



FltEnv



Voice

10.3 Playing it

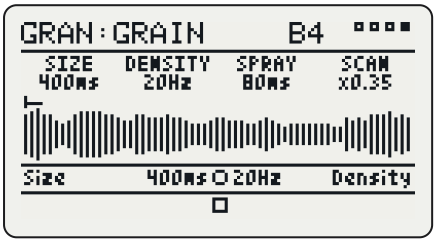
- **Velocity** (MIDI page, B2 then K1): harder than the JP's standard velocity of 80 is louder, brighter (the cutoff opens and the filter envelope sweeps further) and snappier. Softer is quieter, darker and slower to swell.
- **The LFO** is one for the whole engine, so a synced one stays on the beat with the looper. Pitch has a Fade for each note.
- **Mono** plays the last key held and goes back to the one still down when you let go. **Legato** doesn't restart the envelopes on an overlapping note. **Portamento** glides over its time, in Poly too.
- User saves go in **SUPERSAW/** .

> **Processor.** At its heaviest (ten voices sounding, full Detune and Mix, OSC2, 24 dB filter, pitch and filter LFOs) the Supersaw takes about 18–19% of the processor. Everyday sounds take less.

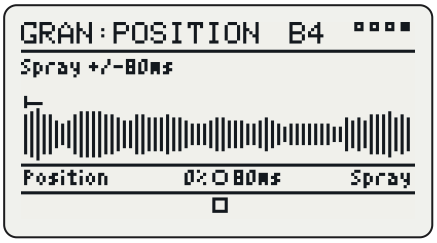
CHAPTER 11

Grains

Grains is a granular instrument. It plays a sound as a cloud of short grains, so it can stretch, freeze, scatter and re-pitch it. The sound can be captured live, taken from an audio track, imported as a WAV, or loaded with a preset. Open it from **Global:Grains**.



Grain, with Airy Piano loaded



Position

11.1 How it plays

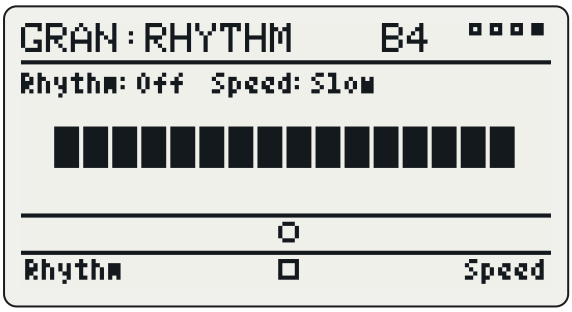
- **Position** is where each note's play pointer sits, and **Scan** moves it through the sound: ×1.00 replays it at its own speed, slower stretches it without changing pitch, negative runs backwards, and Hold keeps it still. Every note starts at Position, and turning Position while notes play carries the whole cloud with it.
- The pointer stays between **Loop Start** and **Loop End**, and Mode sets what it does at the edges.
- **Spray** scatters each grain's start around the pointer. **Direction** is which way each grain plays its own slice, separate from which way Scan moves the pointer.
- **Grain Size** and **Density** set how long each grain is and how many start each second. A note can have up to 32 grains sounding (fewer each in a chord), so very long grains at a high density thin out and get quieter: keep Size in seconds × Density under about 32.
- Up to **four notes** at once, or Mono. With **Keymap** on, each grain plays at the note's pitch; off, every note plays at the Pitch setting. A fifth note takes over the oldest, its grains fading out.

The waveform shows Position's span and a tick for each sounding note's pointer.

11.2 The pages

Page	K1	K2	B1	B2
Grain	Grain Size, 1–1000 ms	Density, 0.5–500 per second		
Position	Position	Spray, ±0–2000 ms		
Scan	Scan, −2× … Hold … +2×		Loop, One shot, Ping pong	Forward, Reverse, Random

Rhythm			Rhythm pattern	Slow, Medium, Fast, Sync
Loop	Loop Start	Loop End		
Pitch	Pitch, ±24 semitones		Keymap on/off	Mono / Poly
ADSR	Attack / Sustain	Decay / Release	Knobs → A + D	Knobs → S + R
Filter, Mix, FX, MIDI	As on every engine (Chapter 7).			
Capture			Source: Direct, Layer, Import	Hold: record or capture
Trim	Start	End	Play the trimmed part	Hold: save the capture
Preset	Starter and Captured folders	Sounds in the folder	Back	Open / load
User	Your saves (Chapter 7).			



Rhythm

Plays the grains in a 16-step pattern instead of a steady stream; B1 picks the pattern and B2 its speed, with Sync following the tempo. The page draws the pattern.

11.3 Capturing a sound

On the **Capture** page, B1 picks where the sound comes from: **Direct** records the input, **Layer** takes an audio track, and **Import** loads a WAV from **GRAINS IMPORT/** on the card. Hold B2 to record or capture. Then on **Trim**, set the start and end with the knobs, play the result with B1, and hold B2 to save it: it goes into Preset → Captured. LED 2 is red while a capture records.

Grains presets keep their sound with them, so a User save brings back the audio as well as the settings. The card template comes with starter sounds in **GRAINS/PRESETS/**, Airy Piano first.

CHAPTER 12

The samplers

Two sampler engines share most of their pages. **SMPVO** (Sample Sound) plays one sound across the keyboard: a single sample, a multi-sample, or a capture. **SMPDR** (Sample Drum) plays a kit or a break, one sample per key from C2. Open them from **Global:SMPVO** and **Global:SMPDR**.

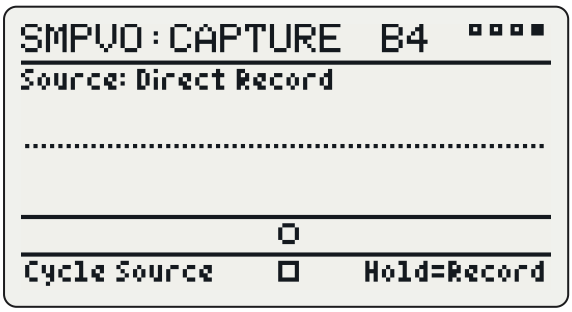
12.1 Pages both have

Page	K1	K2	B1	B2
Filter	Cutoff	Resonance	Filter type	
Env	Attack / Sustain	Decay / Release	Knobs → A + D	Knobs → S + R
Mix	Level / Gain	Pan	Knobs → Level + Pan	Knobs → Gain
FX	Reverb / delay send	Reverb size / delay time	Reverb	Delay
MIDI	MIDI channel	Velocity: how much softer notes play quieter		
Import	Destination		Back	Hold: import
Preset	Types, then folders	Items in the folder	Back	Open / load
User	Your saves (Chapter 7).			

Velocity starts at 50% on SMPVO and at 0% on SMPDR, where every hit plays at full level until you turn it up. It's kept with each User save.

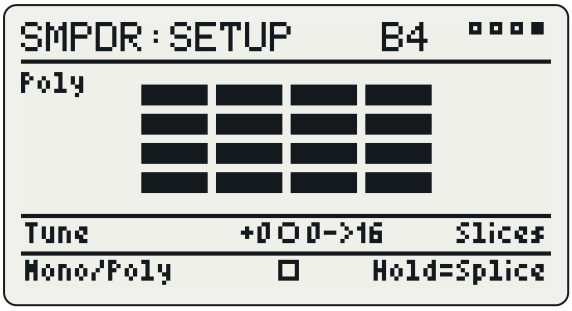
12.2 SMPVO: one sound across the keyboard

SMPVO's Preset page starts with **Sounds**, **Captured** and **Multi**, then a folder, then an item. Its own pages capture and tune a sound:



Capture: B1 picks the source (Direct, Layer or Import) and holding B2 records or captures. **Trim**: K1 and K2 set the start and end, B1 plays the result and holding B2 saves it as a Sound. **Tune**: K1 tunes captures and K2 imports, ±24 semitones, and B1 and B2 switch each between Mono and Poly.

12.3 SMPDR: kits and breaks



SMPDR's Preset page goes straight to its **Kits** and **Breaks**. A kit plays its pads poly; a break starts with Choke on, so each new pad cuts the last.

Setup: K1 tunes the kit, ±24 semitones. B1 switches Mono (choke) and Poly for this kit. K2 sets how many slices to cut a single loop into, 0–16, and holding B2 applies it.

12.4 Sounds, kits and breaks from a computer

The sample library lives in numbered folders of up to 32 items under **SAMPLES/** on the card: **SOUNDS1**, **CAPTURED1**, **MULTISMP1**, **KITS1**, **BREAKS1** and so on. When one fills, the next is made. You can add to it on a computer, no import needed:

- **Sounds**: copy **.wav** files into a **SAMPLES/SOUNDS** folder. Any sample rate from 8 to 192 kHz, 16 or 24-bit, 32-bit or float, mono or stereo. A note at the end of the name, like **Cello C3.wav**, sets its root.
- **Multi-samples**: make a folder in a **MULTISMP** folder with one WAV per recorded note, the note in each name. Each key plays from the nearest recording. Up to 24 notes and about 33 seconds.
- **Kits and breaks**: make a folder in a **KITS** or **BREAKS** folder and copy WAVs in. They load in name order, one per key from C2 (**01 kick.wav**, **02 snare.wav** …). Up to 24 pads and about 33 seconds.

12.5 Importing OP-1 packs

OP-1 sample packs, as **.aif** files from op1.fun or the OP-1 pack builder, import directly, with their slice points and pitch:

1. Copy the files into the staging folders at the card's root: **DRUM PACK IMPORT/KITS/** for drum packs, **DRUM PACK IMPORT/BREAKS/** for breaks and loops, and **SOUND PACK IMPORT/** for sampler patches.
2. On SMPDR's or SMPVO's **Import** page (on SMPDR, K1 picks Kits or Breaks and B2 opens it), K1 picks the destination: a new bank, or an existing one with room. Hold B2 to import every file in the folder.
3. Imported files move into the folder's **Done/**. A file that fails stays for the next try.

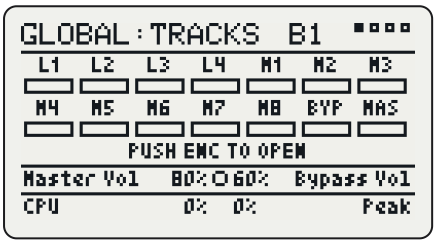
op1.fun often names downloads **patch.aif**: rename it first, to give the kit a proper name.

CHAPTER 13

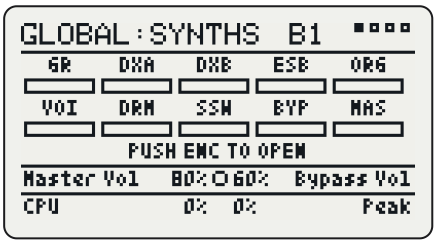
Mixer and effects

Everything meets in the mixer, then the shared effects: one plate reverb and one stereo delay that every track and engine sends into, and a master filter. A soft limiter on the master output catches loud peaks instead of clipping.

13.1 Global:Tracks and Global:Synths



Global:Tracks



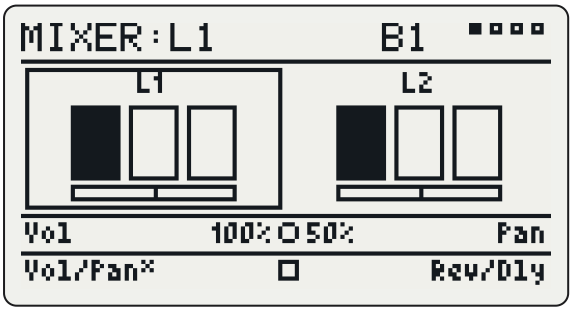
Global:Synths

The first two Global pages are the mixer. **Tracks** shows the four audio tracks and the eight MIDI tracks (L1–L4, M1–M8); **Synths** shows the eight engines. Both end with Bypass (the monitored input) and Master. Each channel is a name over a live peak meter, from −48 dB to 0 dBFS, with a clip mark.

On both, K1 is the **Master Vol** and K2 the **Bypass Vol**, so you can pull a live mic down without touching the mix (Bypass starts at 60%). The bottom row is the live processor load, average and peak; tap B1 for a breakdown by part.

A MIDI track has no sound of its own, so its channel *is* its engine's: M1 playing DXD-A shows and sets DXD-A's level, and turning one turns the other.

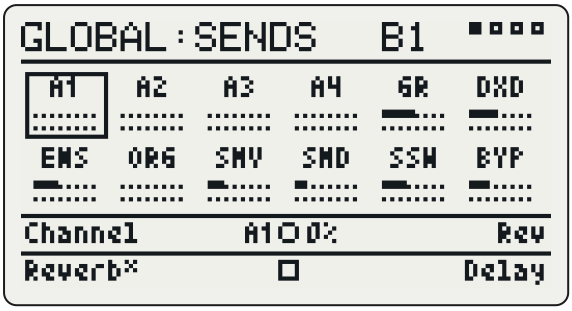
A channel's settings



Click the encoder to step into the channels one by one; turn to move between them, and past the last for a live oscilloscope of the final mix. On a channel, K1 is its volume and K2 its pan. B2 (**Rev/Dly**) turns them into its reverb send (K1) and delay send (K2), and B1 brings back volume and pan. On Master, K1 is the volume and K2 the reverb's size. A MIDI track's channel is titled with its engine, like `M1>DXA`.

Every channel pans with an equal-power law, −3 dB in the centre.

13.2 Global:Sends

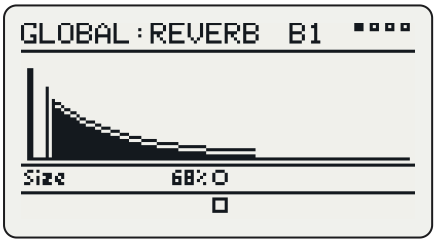


Every channel's reverb and delay send on one page: the audio tracks, each engine (DXD-A and DXD-B share the Dexed bus's) and Bypass, each with two bars. K1 picks the channel; B1 points K2 at its reverb send and B2 at its delay send.

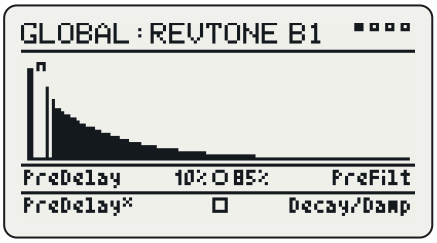
13.3 The reverb

One shared plate reverb, in the Dattorro design. Its pages draw a decay graph: the pre-delay gap, the early reflections and the tail.

Page	K1	K2	B1	B2
Reverb	Size (the decay)			
RevTone	PreDelay / Decay Diffusion	PreFilter / Damping	Knobs → PreDelay + PreFilter	Knobs → Diffusion + Damping
RevDiff	Input Diffusion 1	Input Diffusion 2		



Global:Reverb



Global:RevTone

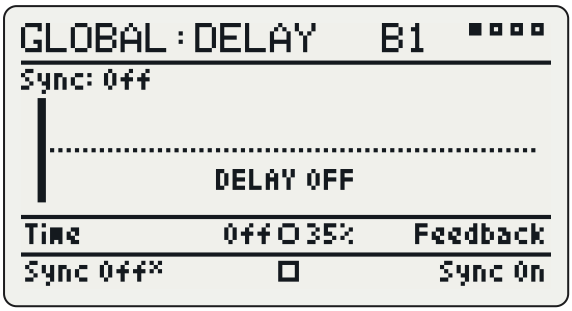
On the graph, the tail's length is Size, its height PreFilter, its texture Decay Diffusion, and its thinning towards the end Damping.

13.4 The delay

One shared stereo delay, with free time or a note value synced to the tempo, ping-pong, a tone that darkens each repeat, and a send that puts its echoes into the reverb. Its pages draw an echo timeline.

Page	K1	K2	B1	B2
Dly-Rev	Delay → Reverb			
Delay	Time: Off, 20 ms–3 s, or a note value	Feedback	Sync off (free time)	Sync on (1/16 … 1/2)

DlyTone	Ping-Pong	Tone (darker repeats)



The timeline's spacing is Time, shrinking by Feedback, with ping-pong above and below the line, thinner repeats for a darker Tone, and a haze after each for Dly-Rev. Beat ticks show when it's synced. At 0% Ping-Pong the echoes follow the sound's pan; at 100% they always bounce left and right.

13.5 The master filter and Bypass

Global:Filter filters the whole mix: K1 cutoff, K2 resonance, B1 the type (Off, low-pass, high-pass, band-pass).

Global:Bypass sets the monitored input's own reverb and delay sends, with the same scope and meter as an engine's Mix page. Switch Bypass on with B2 on Home.

13.6 Global:Speed

Vari-speed for the whole project: K1 changes the speed and pitch of everything together, and B2 resets it to 1.0×. B1 switches the transport between **Normal**, **Scrub** and **Freeze**. In Scrub, the encoder moves through the loop like a tape; in Freeze, it nudges the frozen spot. Hold the encoder to leave.

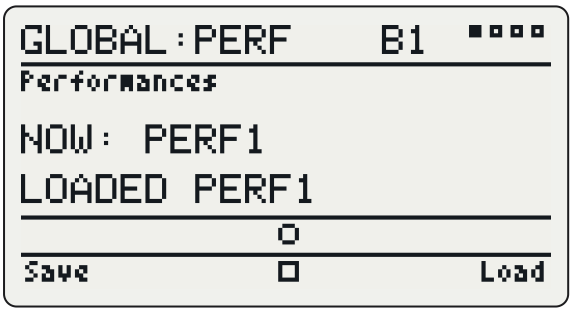
CHAPTER 14

Saving, export and the card

Nothing saves by itself: everything reaches the card when you save it. There are three kinds of save, plus each engine's own User saves (Chapter 7).

Save	Keeps	Page
Performance	Everything: the audio tracks, the MIDI tracks with their patterns and chains, the scenes, the song, every engine's sound, and the looper and global settings.	**Global:Perf**
Snapshot	Every engine's sound and settings at once, which engines are on, and their MIDI channels. For bringing a set of sounds into a new song.	**Global:Snap**
Defaults	What the Groovebox starts with: tempo and mix (Chapter 3), MIDI channels (Chapter 15), the screen saver, the looper on or off.	Each setting's page

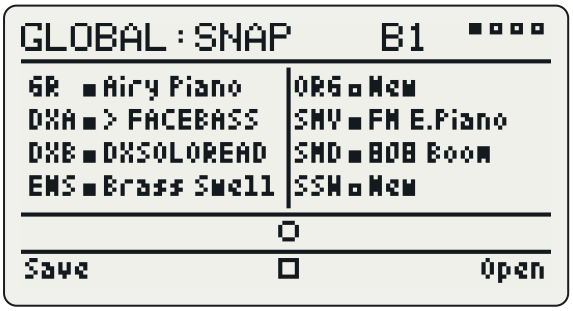
14.1 Performances



On **Global:Perf**, tap B1 to save: K1 picks **Overwrite** (the Performance loaded) or **Save New**, and holding B2 confirms. To load, tap B2 to open the list, turn K2 and tap B2. The list starts with **New (blank)**. A ***** after the title means something has changed since the last load or save.

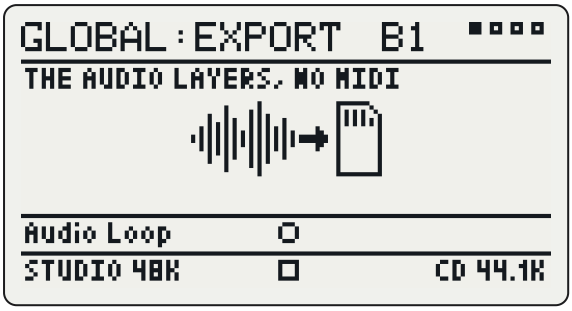
The card template's **PERF001** is the demo song.

14.2 Snapshots



Global:Snap saves the same way: B1, K1 for Overwrite or Save New, hold B2. B2 opens the list, K2 scrolls and B2 loads. The page lists the engines in two columns, each with its on/off box (filled = on) and its sound's name: what's live now, or while browsing, what the highlighted snapshot holds. Grains' audio is saved in the snapshot; the samplers' sounds and kits are saved by name.

14.3 Export

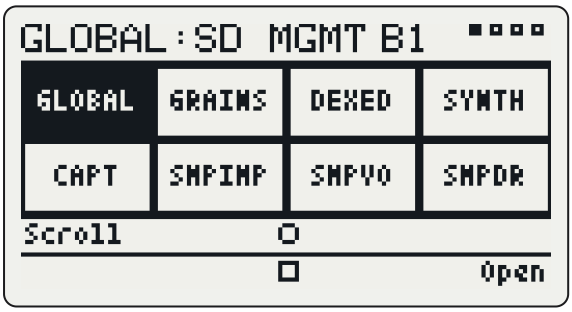


Global:Export writes a WAV to the card. K1 picks what:

- **Audio Loop**: one loop of the audio tracks, rendered in a few seconds and normalised. No MIDI tracks.
- **Audio+MIDI Loop**: everything you hear except the click, recorded live from bar 1 for 1 to 8 loops (K2), up to about 38 seconds.
- **Song**: the song, from the step picked on the Song page to its end (Chapter 5).

Tap B1 for a full-quality 48 kHz file, or B2 for 44.1 kHz, saved straight into a **custom/** folder so the same card works in a MicroDexed Touch's second slot. Tap either again to stop. In the two live modes the master volume sets the level, with the limiter catching peaks.

14.4 Managing files on the card



Global:SD Mgmt copies and deletes files without a computer. It opens on a grid of the eight kinds of save: Global (Performances or Snapshots), Grains and Dexed User saves, SYNTH (the Ensemble's, Organ's or Supersaw's), Captures, Sampler imports, and the SMPVO and SMPDR User saves.

K1 moves around the grid and folders and B2 opens. In a file list, K2 scrolls; B1 goes back, or held, duplicates the file; holding B2 for 1.5 seconds deletes it.

14.5 The card's folders

Folder	Holds
PERFORMANCES	Performances (PERF001.DAT is the demo).
SNAPSHOTS	Snapshots.
DEXED	Dexed User saves and imported banks.
ENSEMBLE, ORGAN, SUPERSAW	Those engines' User saves.
GRAINS	Grains presets, with their audio.
SAMPLES	The samplers' Sounds, Captures, Multi-samples, Kits and Breaks.
EXPORTS	Exported WAVs.

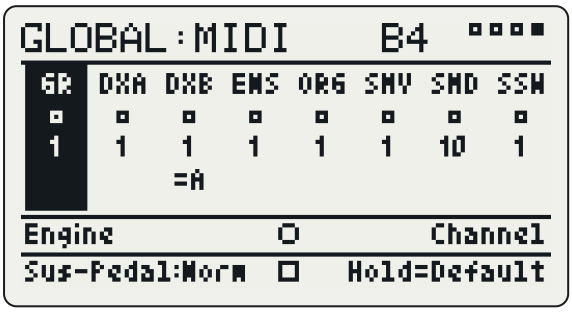
custom	44.1 kHz exports for MicroDexed Touch.
DEXED IMPORT, GRAINS IMPORT	Files waiting to import: DX7 banks, Grains WAVs.
DRUM PACK IMPORT, SOUND PACK IMPORT	OP-1 packs waiting to import (Chapter 12).

Most folders are made the first time you save something into them; the template makes the import folders, which aren't.

CHAPTER 15

MIDI and reference

15.1 Global:MIDI



Every engine's MIDI channel in one place, one column each: GR, DXA, DXB, ENS, ORG, SMV, SMD and SSW, each with its on/off dot and channel. K1 picks an engine and K2 sets its channel, Omni or 1–16: the same setting as the engine's own MIDI page. Hold B2 to save every channel as its power-on default.

While DXD-B is locked to DXD-A it shows **=A**; giving it a different channel here unlocks it.

15.2 The sustain pedal

A pedal sending CC 64 holds notes on every engine, each on its own MIDI channel. Letting go releases every held note except keys still down, and playing a held note again restarts it. Sustained notes count towards an engine's voices.

If your pedal works backwards (some keyboards detect the pedal at power-on), tap B1 on **Global:MIDI** to switch between **Sus-Pedal:Norm** and **Sus-Pedal:Rev**; it's saved straight away. A **SUS** box in the page's title shows the pedal live, filled while it's down.

15.3 The mod wheel

The Ensemble, Organ and Supersaw each pick what the mod wheel moves, with B1 on their MIDI page: the Ensemble and Organ Off, Chorus, Level or Filter; the Supersaw Off, Detune, Cutoff or Level.

15.4 Stuck notes

Triple-click the encoder, or send CC 123 (All Notes Off): every engine lets go of every note, sustained ones too. **ALL NOTES OFF** shows briefly.

15.5 Keeping the processor in hand

The Groovebox runs at 480 MHz, and every engine skips its work while it's silent, so an engine that's on but not playing costs almost nothing. With every engine playing chords over four looping tracks with effects, the load peaks in the mid-80s. To keep headroom:

- Switch off engines a song doesn't use: off frees their share entirely.
- Lower an engine's **Voices** on its MIDI page, especially Dexed's, when several engines play. The Organ is the cheapest of the chord engines.
- Record synth parts into audio tracks, then switch those engines off.
- Set the reverb and delay last: several engines sending into the effects at once is the heaviest load.
- A track's Drive and Tremolo cost nothing at 0.

The bottom row of **Global:Tracks** and **Global:Synths** shows the load live; tap B1 there for a breakdown: Grains, Dexed, Ensemble, Organ, SMPVO, SMPDR, Supersaw, the looper, the delay, the reverb and everything else, each as average and peak.

15.6 If something isn't right

What happens	Try
A note won't stop	Triple-click the encoder.
No sound from a keyboard	Is the engine ENABLED on its Global page? Is the keyboard on the engine's channel (1 to start with)? Check Master Vol (K1 on Home), the engine's Mix level and the headphone wheel.
The keyboard does nothing	Check the cable is Type A and plugged into MIDI In. MIDI doesn't come in over USB.
The screen is dark	The screen saver: touch any control to wake it. That first touch does nothing else.
"No SD card"	Switch off, push the card fully in, and switch on again.
Tempo or bars won't change	They're locked while an audio track holds audio (**LOCKED** on Global:Tempo). Clear the tracks first.
New firmware didn't install	A Pod new to this firmware needs the bootloader once (Chapter 1).